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PROGRESS IN THE PREDICTION, MEASUREMENT AND SIMULATION OF MINERAL LIBERATION

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INTRODUCTION

The importance of mineral liberation has always been recognized but theoretical and experimental methods to deal with the problem have not been easy to develop. In spite of a substantial effort by Gaudin in 1930's to define the problem and to describe some of its practical implications, the next thirty years saw virtually no progress in either theoretical or experimental methods. However, during that time important foundations were being laid by mineralogists who were, at least, devising qualitative schemes for the description of mineralogical texture. It was not until the latter 1960's that renewed interest in the mineral liberation problem prompted Wiegel in the United States and Bodziony in Poland to consider the possibility of developing practical computational schemes that recognized the essentially stochastic nature of the problem and that could be used to quantify mineral liberation and, therefore, make it available as a quantitative tool in practical mineralogical and metallurgical investigations. At about the same time, Jones at Imperial College in London, initiated what was to be a profoundly important experimental study that developed the method of linear intercept analysis into a powerful quantitative laboratory method. It is probably not fortuitous that the population balance method was emerging, at that time, as powerful technique that could be used for the modeling and simulation of comminution and mineral processing unit operations and flow sheets. The work of Reid in Australia on the application of population balance to comminution and of Zaidenberg in Russia, Inoue and Imaizumi in Japan, Woodburn and Loveday in South Africa on flotation modeling exposed the need for liberation analysis if population balance methods were to provide successful and useful methods of practical mineral processing calculations. However it was not until comparatively recently that liberation analysis has been incorporated into the population balance models. The intimate interaction between population balance methods and mineral liberation will be examined in some detail later in this paper. This renewed interest in liberation analysis triggered a concerted research endeavor that has turned out to be remarkably successful and the past 30 years has seen considerable advance in the understanding of mineral liberation - how it occurs, how mineral liberation analysis can be measured and how it can be effectively used in day-to-day operations in mineral processing. Mineral liberation analysis is now taking its rightful place among the standard, experimental and analytical methods in the mining and metallurgical industries.

When discussing mineral liberation and the role that it plays, it is convenient to recognize three broad classes of problem: the Prediction Problem, the Measurement Problem and the Simulation or Comminution problem. There is some overlap among these problem classes but each has generated specific solutions and it is convenient to discuss them separately.

PREDICTION OF MINERAL LIBERATION

The prediction problem centers around the ability to use information about the mineralogical texture in an ore body to predict the liberation characteristics of the ore when it is subject to crushing and grinding only. The essential assumption is that the entire ore sample is comminuted and the liberation spectrum is required in the entire product. There is no selective removal of particles so that the original mineralogical texture and the resulting particles both form space-filling geometrical structures. The resulting liberation spectra are usually required on a size-by-size basis. This can be regarded as the classic liberation problem and while its theoretical and conceptual importance cannot be denied, it is not a problem of much concern in the day-to-day operation of mineral processing plants. However, its solution does have application. For example, at the early feasibility stage in the analysis of an ore deposit, it is useful to have some quantitative assessment of the liberation characteristics of the ore and indeed exploration companies depend on an assessment of the idealized grade recovery or washability analysis of an ore using sample material taken from exploratory drill cores. While such information does not predict the actual grade recovery relationships that may be achievable in a practical working concentration plant, it does define the extreme limits of the mineral recovery potential as a function of the overall fineness of grind. These are important concepts in developing the feasibility for economic exploitation of an ore body.

Aside from its applicability to practical problems, theoretical predictions have set the stage for an attack on the far more difficult problem associated with liberation in practical milling circuits and a brief overview of the prediction methods will now be given. Several different approaches have been used to attack the liberation prediction problem and these are reviewed briefly here.

The liberation prediction problem is stated as

> "Is it possible to characterize the mineralogical texture of the unbroken ore in a quantitative way so that the composition spectrum of the comminuted particles can be predicted as a function of particle size?"

this is a difficult problem and the solutions that have been developed are interesting, imaginative and very useful.

The three approaches are identified with the names of the chief protagonists for each, but many researchers have contributed to the developments that are summarized here.

The chief difficulty in solving the basic problem is due to the indeterminate nature of the geometry of the texture and the indeterminate nature of the geometry of the fracture pattern. Mineralogical textures cannot be described in terms of regular geometric elements such as spheres, cubes, etc. and real fracture surfaces are never planar so that the particles produced are very irregular in shape. The first requirement in finding solutions is a useful geometrical characterizatiuon of the mineralogical texture of the ore. This is by no means a simple task. Any useful characterization must include the “shape characteristics” of the mineral phases and also, importantly, their associations if more than one significant mineral is present.

All solution methods adopt the same strategy to handle this problem: the indeterminate geometry of the real world is transformed to a new geometric space which is either determinate or in which the indeterminate geometry can be adequately handled by appropriate mathematical techniques. These different geometrical spaces are represented in the cartoon as different imaginary worlds that are appropriately characterized. Transformations from the real world to the imaginary worlds are represented by three broad arrows spreading outward from the stylized mineral texture in the real world to each of the transformed geometric spaces. The comminution operation is then represented by an appropriate fracture process in the transformed geometrical space and the liberation spectrum in the transformed space is predicted. The transformed spectrum is then transformed back to the real world. The comminution step in each space is represented by a broad arrow marked "Comminution" and the calculation of the theoretical spectrum is identified by a thin arrow with a bold head which is labeled with a brief description of the method used. The transformation back to the real world is represented by the set of three broad arrows pointing back to the real particles in the real world. The individual steps for the three methods are described briefly below.

Meloy's World

The transformation of the real texture to a regular texture is achieved by means of a textural transformation (Meloy, 1990). The real solid is transformed into a regular geometry while conserving appropriate characteristics such as phase, volume or interphase area. The regular geometry and the conserved characteristics are arbitrary and are chosen to make the subsequent comminution operation and spectrum calculation easy. The comminution operation is simulated by breaking the regular solid into simple geometrical objects which may or may not be space filling. The liberation spectrum is easily calculated by applying standard principles of geometric probability which is a well-established, although not well-known, mathematical discipline.

A simple textural transform is illustrated in Figure 2. The volume fraction of mineral v_m and the interphase area per unit volume of mineral S_{Vmg} is determined in the original texture by image analysis. The textural transform defines the new solid geometry as a simple bar having a square cross-section consisting of two phases with a planar interface. Two invariants are imposed on the transform: volume fraction and interphase area per unit volume of mineral phase must be conserved. These fix the lengths of the two mineral mineral segments in the the new solids as follows.

$$\frac{L_{\Phi}}{L_{\Sigma}} = v_{\Phi}$$

$$\frac{1}{L_{\Phi}} = S_{\acute{\alpha}}^{\Upsilon\varepsilon}$$

The fractional liberation of each phase at particle size D follows immediately from simple geometrical probability arguments.

$$\mathfrak{L}_{\Phi}^{\gamma I \Delta} = \frac{L_{\Phi} - 0.5D}{L_{\Phi}} = 1 - 0.5DS_{\acute{\alpha}}^{\Upsilon\varepsilon}$$

$$\mathcal{L}_{\Sigma}^{\gamma I \Delta} = \frac{L_{\Sigma} - 0.5D}{l_{\Sigma}} = 1 - 0.5Dv_{\Phi}S_{\acute{\alpha}}^{\Upsilon\varepsilon}$$

The internal portion of the spectrum is calculated by assuming the particles are spherical and noting that the distance, z, that a spherical particle penetrates from one phase into the other is uniformly distributed in [0,D].

$$f(z) = \frac{1}{D}$$

The volume of a spherical cap is $z^2(3D - 2z)\pi/6$ and the volumetric grade of mineral in the particle is

$$g_{v} = \frac{z^{\theta}(3D - 2z)}{D^{I}}$$

The density function for g_v is obtained immediately as

$$\begin{aligned} f(g_{v}) &= \frac{f(z)}{\left|\frac{dg_{v}}{dz}\right|} \\ &= \frac{D^{\theta}}{6(Dz - z^{\theta})} \end{aligned}$$

Solving for z in terms of g_v from equation (6) and substituting gives characteristic U-shaped curves for $f(g_v)$ which have been extensively discussed by Meloy's group.

The chief difficulty with this approach is that it is impossible to transform the liberation spectrum back into the real world because of the loss of essential information during the original textural transformation and the result remains in the transformed space. The geometrical elements chosen for the analysis in the transformed space are arbitrary and different choices give widely different results.

Barbery's World

The working space in Barbery's world is defined by Matheron's theory of random closed sets. The transformed geometry is related to the real geometry by the spatial correlation function $C(l)$ which is defined as the probability that two points selected at random in the real ore texture and separated by distance l will be in the same phase. The transformed texture is made up of two phases: one constructed from the union of basic random closed sets and the other occupies the remainder of the space. The transformed structure is required to have the same spatial correlation function as the real texture.

The comminution operation is modeled by fragmenting the model texture into space-filling convex particles of indeterminate shape and size.

The liberation spectrum is calculated only approximately using the first two moments of the distribution. The liberation spectrum cannot be transformed back into the real world and the assumption is made that the transformed geometry is sufficiently close to the real geometry to allow the spectra to be considered identical in the real and transformed spaces.

Barbery and his co-workers have modeled the texture of the ore using the interesting boolean models for 2-phase systems that have been described and made popular by authors such as Serra and Stoyan, Kendall and Mecke. These models, which can be calibrated against measured linear-intercept distributions in the ore, provide a geometrical description of the texture, which in turn can be used to predict how the random- fracture pattern that is induced by comminution will produce particles having compositions ranging from completely liberated gangue to completely liberated mineral with every composition between.

The essential equations for Barbery's procedure are given below.

The essence of this predictive method is the spatial- correlation function which provides a second-order statistical description of the texture. The spatial correlation function for the texture is defined by

$C_{12}(l)$ = probability that two points separated by a distance l in space are in phases 1 and 2 respectively.

This correlation function is related to the distribution of linear intercepts through the two phases in the ore and as a result can be measured using image-analysis techniques. The relationship between $C_{12}(l)$ and the two linear-intercept distributions can be given explicitly only in the Laplace Transform domain by

$$\bar{C}_{\Theta\theta}(s) = \text{Lap trans}\left[C_{\Theta\theta}\right] = \frac{1}{s^{\theta}\left(\mu_{\Theta} + \mu_{\theta}\right)} \left[\frac{\left(1-\bar{f}_{\Theta}\right)\left(1-\bar{f}_{\theta}\right)}{1-\bar{f}_{\Theta}\bar{f}_{\theta}}\right] \tag{8}$$

where $\bar{f}_{\varkappa}$ is the Laplace Transform of the linear intercept distribution density through phase i in the original ore.

This method is based on the approximation that the distribution of compositions of unliberated particles in the population is described by an incomplete beta function

$$p(g) = (1 - \upsilon_{\Theta}\mathscr{L}_{\Theta} - \upsilon_{\varpi}\mathscr{L}_{\varpi}) \; \frac{g^{\alpha-\Theta}(1-g)^{\beta-\Theta}}{B(\alpha,\beta)} \tag{10}$$

where g is the volumetric grade of a particle. $\mathscr{L}_1$ and $\mathscr{L}_0$ represent the fractional liberation of mineral (phase 1) and gangue (phase 0) respectively. $B(\alpha,\beta)$ is the beta function and the parameters α and β are related to the first two moments of the liberation spectrum, n_1, and n_2, by the following equations

$$n_\Theta^{\Phi} = \frac{n_\Theta(1-\mathscr{L}_\eta)}{1-n_\Theta\mathscr{L}_\Theta-(1-n_\Theta)\mathscr{L}_\eta} \tag{11}$$

$$n_\theta^{\Phi} = \frac{n_\theta-n_\Theta\mathscr{L}_\Theta}{1-n_\Theta\mathscr{L}_\Theta-(1-n_\Theta)\mathscr{L}_\varpi} \tag{12}$$

$$n' = \frac{n_\theta^{\Phi}}{n_\Theta^{\Phi}} \tag{13}$$

$$\gamma = \frac{1-n'}{n'-n_\Theta^{\Phi}} \tag{14}$$

$$\alpha = n_\Theta^{\Phi}\gamma \tag{15}$$

$$\beta = (1-n_\Theta^{\Phi})\gamma \tag{16}$$

Equation (11) is essentially a four-parameter representation of the liberation spectrum. By virtue of the subsidiary Eqs. (12) - (17) the four parameters are $\mathscr{L}_0$ and $\mathscr{L}_1$ the fractional liberation of each phase, and n_1 and n_2, the first two moments of the liberation spectrum. The calculation of the liberation spectrum is therefore reduced to the determination of these four parameters. These parameters can in fact be estimated from measurements made on sections of the unbroken ore. All that is needed are the measured linear-intercept distributions through each phase $F_i(l)$ and through the particles $p(l)$. These measurements are easy to make using any good image analyzer.

The first moment of the distribution is just the average volumetric fraction of phase 1

$$n_{\Theta} = \upsilon_{\Theta} = \frac{\mu_{\Theta}}{\mu_{\Theta} + \mu_{\eta}} \tag{17}$$

The second moment of the liberation spectrum is estimated using an equation developed by Davy

$$n_{\theta} = \upsilon_{\Theta} - \frac{4\pi E[V]}{E[V^{\theta}]} \int_{\eta}^{\infty} L^{\theta} P(L) \left(\upsilon_{\Theta} \upsilon_{\eta} - C_{\Theta\Theta}(L) \right) dL \tag{18}$$

where V is the volume of a particle.

The fractional liberation of each of the phases is estimated using an approximate relationship suggested by Barbery

$$\mathscr{L}_{\varkappa} = \left(\mathscr{L}_{\varkappa}^{\gamma\Theta\Delta} \right)^{\iota} \tag{19}$$

where $\mathscr{L}_i^{(1)}$ is the apparent liberation of phase i that would be reported by a one-dimensional probe. The apparent linear liberation can be predicted from the linear-intercept distribution measured on the unbroken ore by a formula derived by King

$$\mathscr{L}_{\varkappa}^{\gamma\Theta\Delta} = 1 - \frac{1}{\mu_{\varkappa}} \int_{\eta}^{\infty} \left(1 - P(l) \right) \left(1 - F_{\varkappa}(l) \right) dl \tag{20}$$

where $F_i(l)$ is the cumulative distribution of linear intercepts through phase i in the unbroken ore.

The sequence of Eqs. (11) - (20) for the calculation of the liberation spectrum was developed by Barbery, and it is appropriate that the procedure should be identified as the Barbery Procedure to recognize his pioneering work.

King's World

King's world is one-dimensional. The real texture is sampled rather than transformed and all manipulations are made on the one-dimensional sample. The real texture is sampled by an IUR line probe by image analysis. The sample texture is modeled as a semi-Markov process.

The linear liberation spectrum is calculated exactly by application of the Takàcs theorem. This is an exact procedure which avoids all assumptions and predicts the liberation spectrum from the measured linear-intercept distributions through each phase in the unbroken ore

Let $p(g|D) = \mathscr{F}[P(g_L|g),P(g_L|D)]$ be the solution to the integral equation

$$P(g_\upsilon|D) = \int_\eta^\Theta P(g_\upsilon|g,D)\,p(g|D)\,dg$$

Here g_L represents the apparent linear grade reported by a single-line probe traverse through a particle and $P(g_L)$ the associated cumulative distribution function. Then the liberation spectrum of the three-dimensional particles is given by

$$p(g) = \mathscr{F}\left[P\left(g_\upsilon|g\right),\left(\upsilon_\Theta H_\Theta + \upsilon_\eta H_\eta\right)\right] \tag{22}$$

where the functions H_1 and H_0 are determined entirely by the linear-intercept distributions of the two phases in the unbroken ore together with the linear-intercept distribution for the particles $p(l|D)$

$$H_\varkappa = \int_\eta^\infty \Phi_\varkappa\left(g_\upsilon|l\right)p(l|D)\,dl \tag{23}$$

The functions $\Phi_i(g_L|l)$ can be evaluated independently of the particle shape or size

$$\Phi_\eta(g_\upsilon|l) \;=\; 1 - \frac{1}{\mu_\eta}\int_{\eta}^{\gamma\Theta-\varepsilon_\upsilon\Delta\varsigma}\left(1 - F_\eta(u) - \sum_{\varphi=\Theta}^{\infty} F_\Theta^{\gamma\varphi\Delta}(g_\upsilon l)\,\Omega_\eta^{\gamma\varphi\Delta}(u)\right) du \tag{24}$$

$$\Phi_\Theta(g_\upsilon|l) \;=\; \frac{1}{\mu_\Theta}\int_{\eta}^{\varepsilon_\upsilon\varsigma}\left(1 - F_\Theta(u) - \sum_{\varphi=\Theta}^{\infty} F_\eta^{\gamma\varphi\Delta}(l - g_\upsilon l)\,\Omega_\Theta^{\gamma\varphi\Delta}(u)\right) du \tag{25}$$

In these equations, $F_i(l)$ is the cumulative distribution of linear intercepts through phase i in the unbroken ore, $F_i^{(n)}(u)$ is the n-fold convolution of $F_i(u)$ with itself and $F_i^0(u) \equiv 1$ and

$$\Omega_\varkappa^{\gamma\varphi\Delta}(u) \;=\; F_\varkappa^{\gamma\varphi-\Theta\Delta}(u) \;-\; 2F_\varkappa^{\gamma\varphi\Delta}(u) \;+\; F_\varkappa^{\gamma\varphi+\Theta\Delta}(u) \tag{26}$$

μ_i is the mean of the distribution F_i.

Although these equations look rather unpromising, they are actually quite easy to evaluate from the measured or theoretical linear-intercept distributions for each of the phases. However the back transformation to the 3-D world by solution of the integral equation (22) to the integral Eq. (2) is not a simple task. The numerical solution is notoriously unstable and requires very careful treatment. In addition, an appropriate kernel $P(g_L|g)$ must be used. At the present time, the best kernels available are those generated by simulated linear-intercept analysis of computer-generated two-phase particles although recent work at the Comminution Center has produced kernels directly from carefully fractionated mineral ores.

The back transformation to the real world is an exact stereological transformation. The strength of King's procedure is that it is mathematically exact at every stage and no approximations are required at any stage including the re-entry to the real world.

Solution of the prediction problem using computer simulation of the texture and the fracture pattern is starting to emerge as a viable and important technique. Simulation of two- or multi-phase mineralogical textures that can be convincingly calibrated back to precise data measured on real textures is now a possibility. The popular Boolean textures based on random set theory have yielded particularly well to quantitative calibration for at least one ore (King 1996). Other simulation models for textures are under development and realistic simulations of typical fracture patterns seem to be well within the capabilities of existing computer systems. Solutions of the prediction problem using simulation will yield an added advantage in that these methods generate simulated particles of multi-phase materials that can be used for further analysis. They can be used to generate stereological correction kernels directly and the simulated particles can be broken further and the resulting particle populations can shed light on these all-important fracture processes and on the structure of the Andrews-Mika diagram. The significance of the Andrews-Mika diagram is discussed later in this paper.

THE MEASUREMENT PROBLEM

Probably the most significant factor that has inhibited the routine use of mineral liberation analysis for so long has been the lack of a convenient and accurate method for the measurement of the liberation spectrum in samples of particulate solids. Apart from the few mineral suites that lend themselves to analysis by dense-liquid fractionation, no suitable measurement techniques was available until the mid 1980's when digital image analysis provided on effective and accurate method that was not sensitive to variation of the density and other physical properties of the minerals. Mineral liberation measurement by image analysis is now routine and the technique is capable of handling the number of samples that are required for comprehensive evaluation of liberation in operating mineral processing plants. These methods have the added benefit of being able to make use of the latest generation of micro analytical instruments. There is a quite a lot of research effort currently devoted to the improvement of the method through the reduction of errors that are inherent in the stereological correction methods and a major need is the development of stereological correction procedures for multi-component systems (those that contain more than two minerals).

These instruments make it possible to generate digital images of sectioned samples of mineral particles in which the different mineral ore identified. Figure 1 shows a typical image of a two-phase mineral system in particulate form. A good image analysis system that is programmed for mineral liberation work will segment an image such as this and will accurately delineate each of the mineral phases as well as the particle sections. A delineated and segmented version of the image in Figure

1 is shown in Figure 2. Straightforward analysis of the latter image will produce a discrete version of the linear or areal grade distribution usually classified into 12 bins for two component ore or 43 bins for 3 component systems. 22 bins are sometimes used for 2-component systems to provide greater resolution of the measured liberation spectrum for binary systems. This will probably become more common as the demand for greater resolution grows.

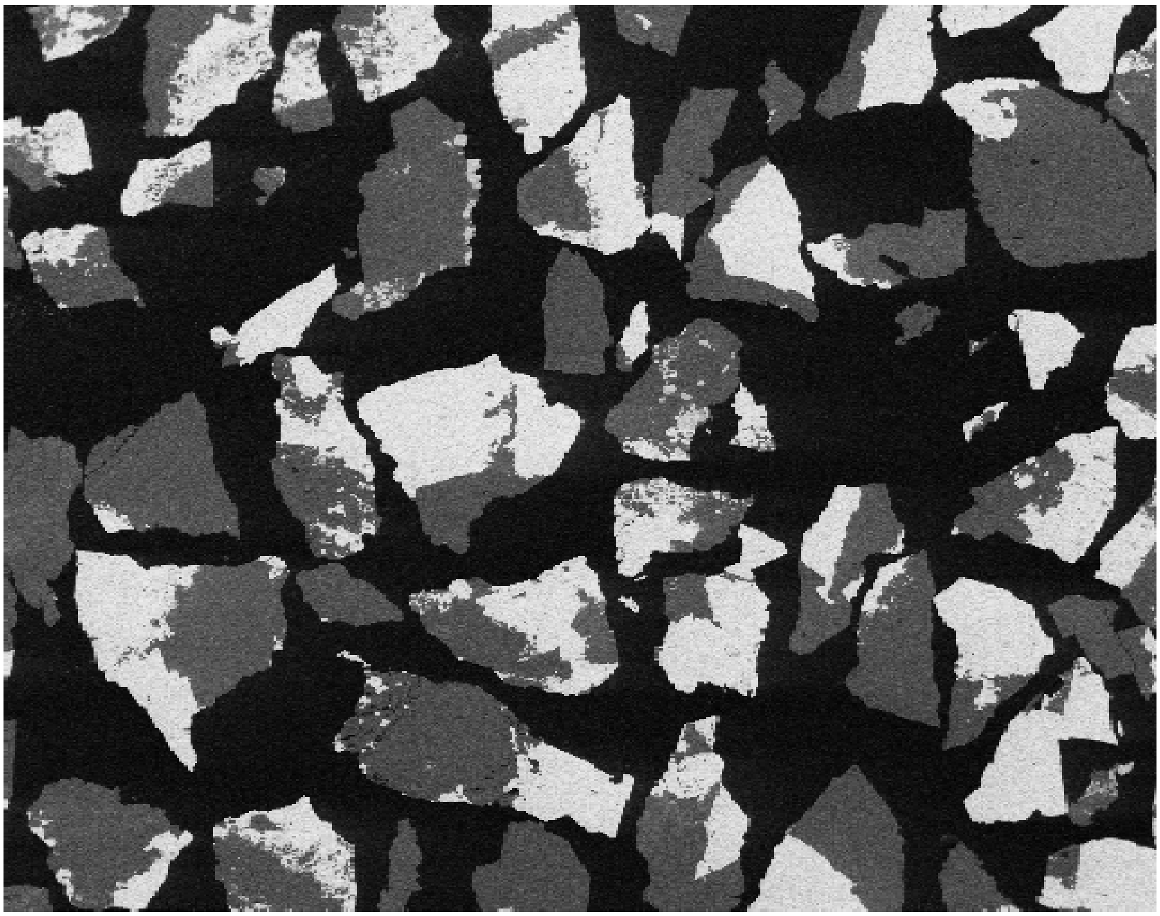

Figure 1 Typical back scattered electron image of two-phase mineral particles.

Modern image analysis systems are capable of generating the sectioned grade distributions with sufficient precision to provide good estimates of the linear and areal grade distributions. However data of this kind must normally be stereologically corrected before it is of any real use for the solution of mineral processing problems. Stereological correction of mineral liberation data is not difficult to do and a variety of algorithms range from simple empirical corrections of the measured section data through to regularized solutions of the associated inverse problem. All of the available algorithms require specialized computing skills but the necessary computer software packages are available. The following brief description of methods that have been used will give the reader some idea of what is available.

Empirical correction method (Barbery 1991, 1992)

This method is based on the assumption that the distribution function for the liberation spectrum can

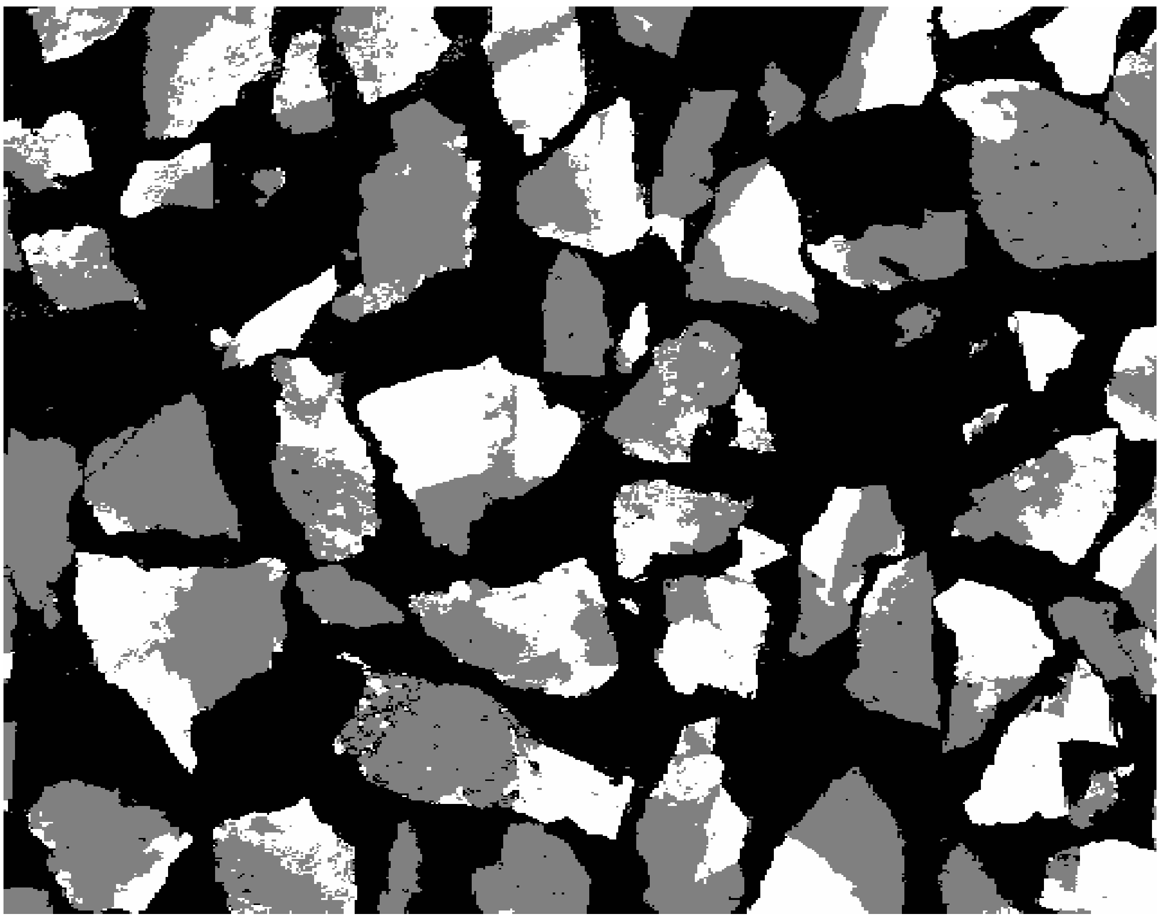

Figure 2 Delineated and segmented version of the image in Figure 1.

be adequately described by the beta distribution. The precise shape of the distribution is determined by its first two moments which are estimated from the experimental section data together with an assumed geometric model for the mineralogical texture in the original ore. The intensity of the spectrum at the two liberated ends are estimated purely empirically from these intensities in the measured section spectrum using

$$\mathscr{L}_{\rho Ⅎ} = (\mathscr{L}_{\rho \Theta})^{\theta} = (\mathscr{L}_{\rho \Theta})^{\iota} \tag{27}$$

Allocation method (Gay and Lyman, 1995)

In this method individual measured sections are allocated to specific composition classes or bins in the 3-dimensional space and the liberation spectrum is generated directly. The allocation seeks to minimize the weighted squared difference between the assumed composition of each bin (usually the midpoint) and the estimated average composition of the allocated sections in that bin. This estimate is obtained from an arbitrary number of stereological estimates of the mean volumetric grade in each bin given the sections that have been allocated to the bin. The weighting factors are proportional to the inverse of the estimated variance of the stereological estimate of the mean. The method is naturally regularized because the allocation cannot ever result in a negative quantity in any bin. Noise suppression is achieved by imposing penalties on noisy distributions including bin-to-bin fluctuations in the distribution of purely geometrical properties such as particle shape and size.

Forward correction method (Hill, Rowlands and Finch, 1987)

This method is based on the premise that all of the stereological error is confined to over estimation of the liberated sections and consequently only the two end classes of the liberation spectrum are corrected.

This method depends on the stereological transformation equation

$$f_{\Upsilon} = K f_{\cdot} \tag{28}$$

to make approximate corrections to the liberated ends of the distribution using the uncorrected inner distribution as an approximation to the true 3 dimensional distribution.

Let

$f_{\cdot}^{\Upsilon}$ represent the true inner distribution

and

K^{Υ} the truncated $N\ x\ N\!-\!2$ inner transformation kernel. K^{Υ} is generated by stripping the first and last columns from K.

Corrected liberated ends are calculated from

$$(f_{\cdot})_{\varkappa} = (f_{\Upsilon})_{\varkappa} - \left(\frac{K^{\gamma\Upsilon\Delta} f_{\cdot}^{\gamma\Upsilon\Delta}}{\left| K^{\gamma\Upsilon\Delta} f_{\cdot}^{\gamma\Upsilon\Delta} \right|} \right) \quad for\ \ i = 1{,}12 \tag{29}$$

This method has no noise control feature and no regularization is applied. Consequently negative values for $(f_{\cdot})_{\varkappa}$ are often found with real data.

Inverse correction method (Coleman, 1988, King and Schneider 1998)

This method is based on the solution of the integral equation for stereological correction.

$$F(g_{\Upsilon}) = \int_{\circ}^{\shortmid} F(g_{\Upsilon} | g_{\cdot}, D)\ f(g_{\cdot})\ dg_{\cdot} \tag{30}$$

This inverse problem is solved using a finite difference approximation

$$F_{\Upsilon} = Kf_{\cdot} \tag{31}$$

using a regularization method proposed by Coleman (1988).

$$Minimize\ \ \| Kf_{\cdot} - F_{\Upsilon} \| + \lambda \Sigma (f_{\cdot})_{\varkappa} \log\ (f_{\cdot})_{\varkappa} \tag{32}$$

Subject to $\Sigma(f_{\cdot})_{\varkappa} = 1$

and

$$(f_{\cdot})_{\varkappa} \geq 0 \;\; for \;\; all \;\; i \tag{33}$$

This method has been tested against known true liberation spectra determined using serial sectioning, elutriation and dense and magnetic fluid fractionation. It has proved to be accurate and reliable in practice. Details of the method are available in King and Schneider (1997, 1998)

THE SIMULATION PROBLEM

The methods of liberation analysis are perhaps most useful when liberation modeling is incorporated into population balance simulations of milling circuits. Methods for doing this have evolved over the past 15 years and efficient and versatile systems are now available. Whenever these have been tested against plant operating data, the results have been good and many applications can be expected during the next few years. The key to the success of the population balance simulations has been the modeling of the Andrews-Mika diagram but much remains to be done in this area. At present very little is known about the relationship between the structure (both the boundary structure and the internal structure)of the A-M diagram and the mineralogical texture of the ore.. For example it is known that grain-boundary fracture influences the A-M diagram but none of the details have yet been worked out. The development of the A-M diagram for multi-component mineral suites is a pressing research problem which is likely to command a lot of attention in the immediate future.

It is impossible to over emphasize the importance of the Andrews-Mika diagram. The information that it contains establishes the essential link between the mineralogical texture of the ore and the disposition of the individual mineral phases in the progeny particles when a single multiphase particle is broken in a comminution machine. The Andrews-Mika diagram is a highly structurede graphical representation of the complex interaction between the mineral phase geometry of the original particle and the fracture surfaces that are created when it is broken. The complete diagram is defined by its boundaries and its internal structure.

The boundaries of the Andrews-Mika diagram were identified and described by Andrews and Mika

(1975) and more recent investigations have revealed quite a lot about the boundaries and their relationship to the liberation process. The internal structure is only now starting to yield to description on the basis of experimental data and this continues to be a fertile field for detailed research.

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